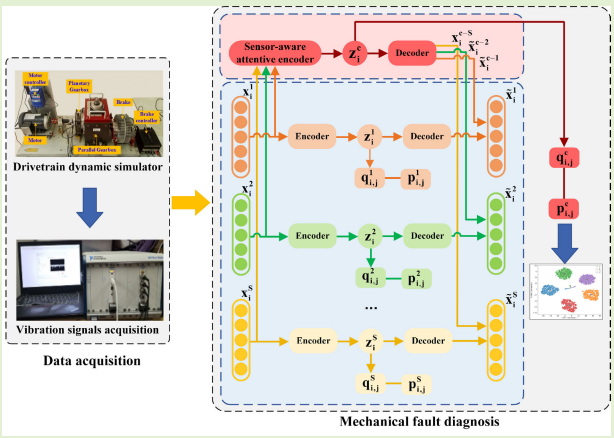
Multisensor Deep Clustering Networks With Joint Learning Strategy for Fault Diagnosis Considering Consensus-Specific Constraint

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***Abstract*—The deep clustering method has been successfully employed for fault diagnosis (FD) due to its remarkable ability to extract deep representation features, particularly in scenarios where labeled data is unavailable. However, the separate learning strategy utilized in deep clustering-based FD methods leads to suboptimal FD results, and the consensus representation feature present in multisensor industrial data is not fully explored to improve the effectiveness of FD. To address the above problems, a novel deep clustering method named multisensor deep clustering networks (MsDCNs) with joint learning strategy is proposed for FD, which can comprehensively utilize industrial multisensor data and enable joint optimization. First, the Commonality-net uses a sensor-aware attentive encoder (SaAE) to integrate multisensor industrial data and extract the consensus representation feature. Second, the Individuality-net is established to extract the specific representation features of multisensor industrial data. Finally, a joint multisensor deep clustering learning strategy, including consensus-specific constraints, is designed to optimize the networks jointly and learn the consensus representation feature with consistent and complementary information. Extensive experiments on the gearbox and bearing datasets validate the proposed MsDCN. The results indicate that the proposed MsDCN can generate a reliable consensus representation feature to achieve better FD performance than other benchmark methods. Moreover, its capability to enhance multisensor FD with unlabeled data highlights its potential for real-world applications.**



***Index Terms*— Consensus representation feature, consensus-specific constraint, deep clustering, fault diagnosis (FD), joint learning strategy, Kullback–Leibler divergence (KL divergence), multisensor data.**

I. Introduction

FAULT diagnosis (FD) is of paramount importance for prognostic health management (PHM) in modern industrial processes, as it can reduce breakdown costs, enhance operational safety, and prevent casualties [1], [2]. Recent advances in sensing technologies and computational algorithms have driven increased research focus on intelligent supervised FD method, enabling integration with emerging approaches such as digital twin-driven systems, cyclostationarity-based monitoring, and neuro-fuzzy-guided diagnostic frameworks [3], [4], [5], [6]. Nevertheless, it is often difficult and expensive to collect sufficient data labels in real industrial environments [7]. Consequently, the unsupervised FD method has become a key research focus.

The clustering-based FD method, as a form of unsupervised FD method, has demonstrated noticeable potential to accurately cluster industrial data by leveraging massive amounts of unlabeled data [8], [9]. Currently, the clustering-based FD method is classified into two categories, i.e., traditional clustering-based FD method and deep clustering-based FD method [10], [11]. Traditional clustering-based FD method generally consists of two major procedures, i.e., feature

Received 19 March 2025; accepted 3 April 2025. Date of publication 2 May 2025; date of current version 2 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 72431005, Grant 72371096, and Grant 72071062; and in part by the Science Fund for Distinguished Young Scholars of Anhui under Grant 2208085J12. The associate editor coordinating the review of this article and approving it for publication was Dr. Qing Li. *(Corresponding author: Gang Wang.)*

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Digital Object Identifier 10.1109/JSEN.2025.3560460

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extraction and fault pattern recognition [12]. In the feature extraction procedure, hand-crafted features are extracted, such as time-domain features, frequency-domain features, and time-frequency-domain features, which are utilized for FD [13], [14], [15]. In the fault pattern recognition procedure, clustering methods, such as K-means, subspace clustering, and fuzzy C-means clustering (FCM), are applied to identify fault patterns [16], [17], [18]. However, the traditional clustering-based FD method has the defect of depending excessively on hand-crafted features, which is time-consuming and labor-intensive. In addition, the collected industrial data are usually characterized by high dimension and complex nonlinearity, and this method struggles to extract pivotal features from high-dimensional industrial data.

Recently, the deep clustering-based FD method has obtained successful application in FD, since its ability to extract deep representation features from high-dimensional industrial data [19], [20]. The prevailing deep clustering-based FD method mainly consists of two stages, i.e., deep representation feature extraction and clustering. Specifically, unsupervised deep learning methods, such as deep belief network (DBN), autoencoder (AE), and sparse filtering (SF) [21], [22], [23], are employed to extract deep representation features. While diverse clustering techniques, such as K-means, FCM, and Gaussian mixture model (GMM), are applied to diagnose fault patterns based on the extracted deep representation features [24], [25], [26].

In summary, while the deep clustering-based FD method has demonstrated remarkable capabilities, there are two major challenges that need to be addressed. On the one hand, the existing deep clustering-based FD method mainly adopts a separate learning strategy, where feature extraction and clustering are performed independently, inherently preventing the generation of clustering-friendly features, as deep representation features are not co-optimized with cluster structures, directly impacting FD performance. On the other hand, multiple sensors have been widely used in the field of industry, and the industrial data collected from multiple sensors installed at different times and locations contains comprehensive information. For multisensor industrial data, it is important to investigate a consensus representation feature with comprehensive information for FD. However, the consensus representation feature is not fully exploited by the existing method, suggesting that there is potential for further improvement in the performance of unsupervised FD.

Aiming at the problems mentioned above, this article proposes multisensor deep clustering network (MsDCN) with a joint learning strategy, fully exploring the consensus representation feature considering complementary and consistent information in multisensor industrial data for FD. Specifically, the Commonality-net utilizes the sensor-aware attentive encoder (SaAE) to process the industrial data collected by multiple sensors and generates the consensus representation feature and the reconstruction of multisensor industrial data. Then, the Individuality-net extracts the specific representation features and obtains the reconstruction of multisensor industrial data from each sensor, respectively. Finally, a joint multisensor deep clustering learning strategy including consensus-specific constraint is proposed to learn the consensus representation feature considering consistent and complementary information for effective FD. Extensive experiments are conducted to assess the efficacy of the MsDCN, which demonstrates that the MsDCN promotes the performance of unsupervised multisensor FD.

The contributions of this article are summarized as follows.

1) A joint deep clustering framework for multisensor FD is proposed in this article, explicitly addressing two critical limitations in existing methods. To resolve suboptimal feature-cluster alignment caused by the separate learning strategy, the framework introduces a joint learning strategy that co-optimizes deep representation feature extraction and clustering tasks, generating clustering-friendly features. Additionally, to tackle the underutilization of multisensor data, it integrates the Commonality-net with an SaAE and the Individuality-net, extracting consensus representation features with consistent and complementary information. These innovations collectively improve unsupervised FD performance in industrial environments.
2) A novel deep clustering method named MsDCN is proposed for FD. MsDCN leverages the Commonality-net with SaAE to explore the consensus representation feature in multisensor industrial data, and employs the Individuality-net consisting of AEs to extract sensor-specific features. Unlike separate learning strategy, the proposed joint multisensor deep clustering learning strategy jointly optimizes deep representation feature extraction and clustering, ensuring that the extracted features are clustering-friendly and construct a more discriminative cluster space. Moreover, this strategy explicitly considers both complementary and consistent information across multisensor data, allowing the model to learn a more comprehensive and informative representation, thereby improving FD performance.
3) The proposed MsDCN is evaluated on real-world industrial datasets, and the experimental results demonstrate its superior performance in FD when compared to the traditional clustering-based FD method and the latest deep clustering-based FD method.

The rest of this article is organized as follows. In Section II, we introduce the related work on traditional clustering-based FD method, deep clustering-based FD method, and deep clustering method with joint learning strategy. In Section III, we describe the framework of the proposed method MsDCN for FD. In Sections IV and V, the description of the industrial dataset is presented, and the details of the experimental results are analyzed. Finally, Section VI refines the article's conclusions and summarizes the future directions.

II. Related Works

The clustering-based FD method provides an effective way to recognize faults without sufficient labeled industrial data. Generally, the clustering-based FD method can be categorized into the traditional clustering-based method and the deep clustering-based method. The two kinds of methods are discussed in this section, respectively.

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A. Traditional Clustering-Based Fault Diagnosis Method

Feature extraction and fault pattern recognition are two main stages of the traditional clustering-based FD method. In the stage of feature extraction, various signal processing technologies are employed to extract features, such as the time-domain, the frequency-domain, and the time-frequency-domain methods [27]. For instance, Srividya et al. [28] processed time-domain signals and input them into a neural network, and the potential of the proposed method was verified by the experimental results of bearing FD. Ceban et al. [29] analyzed the low-frequency part of the magnetic field spectrum to extract the frequency-domain information, and the results indicated the efficacy and simplicity of the proposed method. Ma et al. [30] utilized a 2-D image representation method to transform the vibration signals to time-frequency images for feature extraction, and the experimental results indicated the effectiveness and robustness of the method with FD application.

In the stage of fault pattern recognition, traditional clustering methods are utilized to identify industrial data to several fault patterns. Traditional clustering methods, such as K-means, subspace clustering, and FCM, have been successfully adopted for FD [31]. For instance, Wan et al. [32] adopted ant colony optimization (ACO)-K-means clustering algorithm to perform clustering for bearing FD, and the accuracy and efficiency of the method were verified by the experimental results. Gao et al. [33] mapped the high-dimensional data into a low-dimensional space and adopted the subspace clustering method to learn latent features for FD. The results indicated the efficacy of the method with bearing FD application. Hou et al. [34] utilized the Chi-square Test algorithm, Variance-Relief-F algorithm, and hierarchical clustering algorithm to obtain the optimal features, and adopted FCM to complete FD. Extensive experiments were conducted to verify the effectiveness and feasibility of the method. However, the traditional clustering-based FD method has the drawback of being time-consuming and labor-intensive due to its over-reliance on hand-crafted features extracted by signal processing technologies. Additionally, the method is unable to extract crucial features from high-dimensional industrial data, which can negatively affect the performance of FD.

B. Deep Clustering-Based Fault Diagnosis Method

Recently, in the face of the above problems caused by the traditional clustering-based FD method, the deep clustering-based FD method has become an effective solution. Due to its remarkable capability in deep representation features learning, the deep clustering-based method has been successfully employed in the field of FD [35]. The method mainly consists of two stages, i.e., deep representation features extraction and clustering. In the stage of deep representation features extraction, deep representation features are extracted by unsupervised deep learning methods. Several unsupervised deep learning methods, such as DBN, AE, and SF, have been successfully applied in FD [36]. For example, Gan et al. [37] proposed a novel hierarchical diagnosis network (HDN) consisting of DBNs that extracted discriminative features for FD, and the experimental results showed the reliability of HDN for precise multistage diagnosis. Xu et al. [38] utilized a stacked denoising AE (SDAE) to extract the deep representation features, and the results showed that the classification accuracy of the FD method was superior to the current advanced methods. He et al. [39] proposed a multiview feature learning network with the SF for gearbox FD, which extracted complementary and consistent features from different views, and the experimental results demonstrated the effectiveness and superiority of the proposed method.

In the stage of clustering, plenty of clustering methods, such as K-means and clustering fast searching (CFS), have been successfully applied to recognize the fault pattern. For example, Li et al. [40] presented a deep clustering-based FD method, in which distance metric learning and K-means clustering method were employed for feature optimization and fault pattern recognition, respectively, and the experimental results verified the superiority of the proposed method for FD. Wang et al. [41] developed an ensemble clustering-based FD method, utilizing the gated recurrent units-based AE for feature extraction and K-means for fault pattern recognition. The experimental results illustrated the effectiveness of the method compared with other existing benchmark clustering methods in unsupervised FD. Xu and Tse [42] employed the CFS algorithm for bearings FD without principal components analysis, and the experiments validated the superiority of the proposed method. However, in the field of FD, the existing deep clustering-based method typically adopts a separate learning strategy, where feature extraction and clustering are independent processes. This separation prevents the optimization of deep features for clustering, resulting in suboptimal FD performance.

C. Deep Clustering Method With Joint Learning Strategy

Recently, the deep clustering method with a joint learning strategy has been proposed. The method can jointly optimize the extraction of deep representation features and the execution of clustering, which has become a popular research topic [43]. From the perspective of the clustering model, the joint deep clustering method mainly includes the K-means-based clustering method and the Kullback–Leibler divergence (KL divergence)-based clustering method. The K-means clustering-based method optimizes feature learning and the traditional K-means clustering method simultaneously. For instance, Yang et al. [44] presented a deep clustering network (DCN) that combined a deep neural network and K-means clustering approach to extract clustering-friendly features, and the experimental results validated the effectiveness of the proposed approach. Fard et al. [45] introduced a deep K-means method (DKM) that clustered with K-means and learned deep representation features by the clustering loss. The experiments demonstrate the exceptional performance of this approach. Ma et al. [46] introduced a deep temporal clustering representation (DTCR) that integrated the temporal reconstruction and K-means objective to attain deep representation features, and the experimental results verified the remarkable performance of DTCR.

Compared with the K-means-based clustering method, the KL divergence-based clustering method is highly efficient and flexible. The KL divergence-based clustering method

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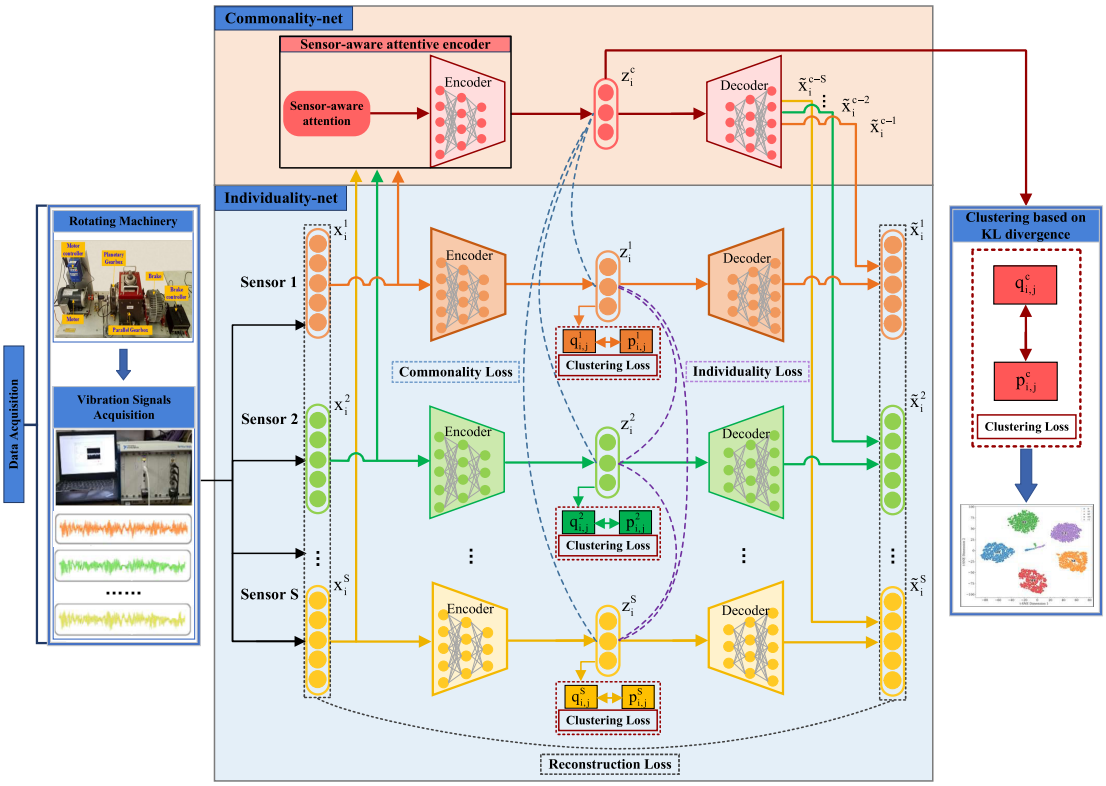


Fig. 1. Framework of the proposed MsDCN.

optimizes the clustering centers by calculating soft distribution and auxiliary target distribution. For instance, Xie et al. [47] proposed deep embedded clustering (DEC), which was the first to minimize KL divergence to an auxiliary target distribution to optimize clustering assignment and feature learning simultaneously. The experimental results validated the improvement of DEC. Guo et al. [48] proposed the improved DEC (IDEC) algorithm to learn deep representation features and accomplish cluster label assignment simultaneously by minimizing reconstruction loss and clustering loss, and the effectiveness of IDEC was indicated by the experiments over state-of-the-art methods. Dizaji et al. [49] introduced a novel clustering model named deep embedded regularized clustering (DEPICT) that mapped data into an embedding subspace and obtained cluster assignments by KL divergence minimization, and the efficiency of DEPICT was validated by experiments on real-world datasets.

In conclusion, the deep clustering-based FD method with a separate learning strategy has been successfully applied in the field of industry, yet there are still two difficulties that degrade FD performance. On the one hand, the existing deep clustering-based FD method utilizes a separate learning strategy instead of a joint learning strategy. This kind of method hinders simultaneous optimization for extracting clustering-friendly features, ultimately leading to suboptimal FD performance. On the other hand, the presence of a common representation feature in multisensor industrial data offers significant potential for precise FD. Nevertheless, the aforementioned methods have yet to fully explore the common representation feature to further improve fault diagnostic performance. Therefore, it is essential to introduce a joint deep clustering method attaching importance to consensus representation feature considering consistent and complementary information for FD.

III. Proposed Method

A. Framework

In this study, we propose MsDCN fully exploit the consensus representation feature to improve FD performance. MsDCN integrates two networks, namely the Commonality-net and the Individuality-net, and its framework is illustrated in Fig. 1. First, the Commonality-net integrates multisensor data by employing an SaAE, extracts the consensus representation feature from the processed data, and decodes it to generate reconstructed multisensor data. Here, the consensus representation feature is explicitly defined as a unified feature learned from multisensor data, capturing the global information across different sensors, thus comprehensively reflecting the overall condition of the industrial equipment Second, the Individuality-net prepares an AE for each sensor, respectively. It extracts the specific representation features for each sensor and obtains the reconstruction. Finally, the joint multisensor deep clustering learning strategy with consensus-specific constraint is proposed to simultaneously optimize the network, meanwhile learning the consensus representation feature considering consistent and complementary information for FD.

For a clear presentation, some notions used in this article are defined. Given a multisensor industrial dataset $D = \{x_i^1, x_i^2, \ldots, x_i^S\}_{i=1}^N$, where $x_i^s \in \mathbb{R}^{d_s}$, S is the total number of sensors, N is the number of instances, and d_s is the dimensionality of sensor. Additionally, z_i^c represents the consensus representation feature, which captures global information across sensors. z_i^s denotes the sensor-specific feature,

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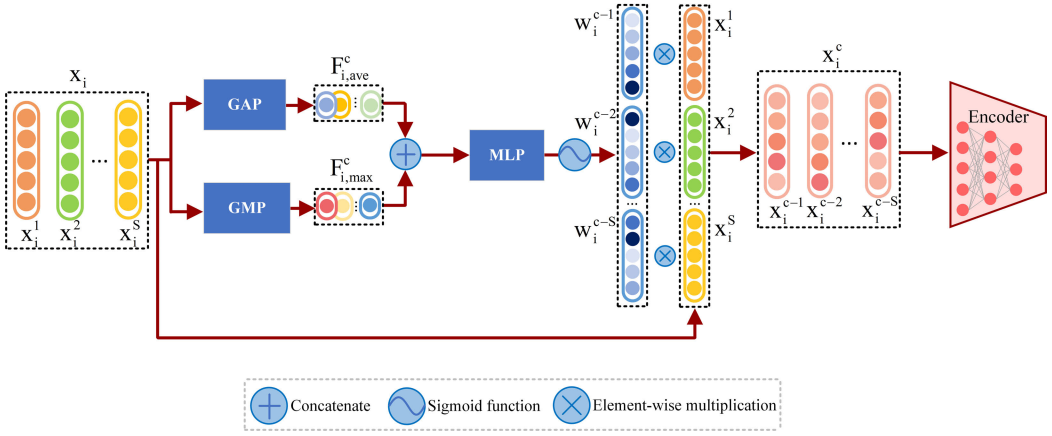


Fig. 2. SaAE.

which captures the local information of each sensor. The goal is to categorize the industrial data into K clusters.

B. Commonality-Net

The Commonality-net extracts a consensus representation feature and obtains the reconstruction to retain consistent information, where AE is utilized for the process of feature extraction and data reconstruction. The reason we chose AE is that the features extracted by AE are both simple and flexible, making them highly suitable for FD. Comprising SaAE, the Commonality-net enables the extraction of consensus representation feature from multisensor industrial data, which facilitates the effective utilization of comprehensive information across multiple sensors. Meanwhile, the decoder layer reconstructs the input multisensor data based on the extracted feature. By emphasizing shared characteristics across different sensor inputs, the Commonality-net improves the overall accuracy of the clustering results.

As shown in Fig. 2, the role of SaAE is to allocate distinct weights to various industrial data of each sensor, predicated upon the relative importance of multisensor industrial data. To achieve this, the SaAE initially utilizes average-pooling and max-pooling operations to process multisensor industrial data x_i^s, generating average-pooled features $F_{i,\text{ave}}^s$ $(s = 1, 2, \ldots, S)$ and max-pooled features $F_{i,\max}^s$ $(s = 1, 2, \ldots, S)$ for each sensor

$$F_{i,\text{ave}}^s = \frac{1}{d_s}\sum_{d=1}^{d_s}\left(x_i^s(d)\right) \tag{1}$$

$$F_{i,\max}^s = \max_{d=1,2,\ldots,d_s}\left(x_i^s(d)\right). \tag{2}$$

Then, the two kinds of features are put together to attain $F_{i,\text{ave}}^c$ and $F_{i,\max}^c$, respectively. Then $F_{i,\text{ave}}^c$ and $F_{i,\max}^c$ are concatenated to feed into a shared multilayer perceptron (MLP) network, which comprises a hidden layer. Following the execution of MLP operations, the attention weights w_i^c are obtained via an activation function set to sigmoid:

$$w_i^c = \sigma\left(w^c\left(F_{i,\text{ave}}^c \oplus F_{i,\max}^c\right) + b^c\right) \tag{3}$$

where $\sigma(\cdot)$ represents the sigmoid function, w^c and b^c represent weight matrix and bias of MLP, respectively, and $\oplus$ denotes the concatenate operation. Finally, the output features are calculated by performing element-wise multiplication

$$x_i^c = w_i^c \otimes x_i \tag{4}$$

where x_i^c is composed of x_i^{c-s}, and $\otimes$ denotes the element-wise multiplication.

The next step of SaAE is to map x_i^c into consensus representation feature

$$z_i^c = f_{\text{en}}^c\left(w_{\text{en}}^c x_i^c + b_{\text{en}}^c\right) \tag{5}$$

where z_i^c is the consensus representation feature, f_{en}^c denotes the activation function of the encoder layer, and w_{en}^c and b_{en}^c denote the weight matrix and bias of the encoder layer, respectively. The decoder layer then converts the consensus representation feature into the reconstruction of the original multisensor industrial data

$$\tilde{x}_i^c = f_{\text{de}}^c\left(w_{\text{de}}^c z_i^c + b_{\text{de}}^c\right) \tag{6}$$

where $\tilde{x}_i^c$ is the reconstruction of the origin multisensor industrial data, including reconstructed industrial data $\tilde{x}_i^{c-s}$ from each sensor, and w_{de}^c and b_{de}^c are the weight matrix and bias of the decoder layer, respectively. The optimization of the parameters of the Commonality-net is achieved through the minimization of the reconstruction error that exists between the original multisensor industrial data and the reconstructed multisensor industrial data. To this end, the mean square error (mse) is utilized as a measure of the average reconstruction error in Commonality-net

$$L_{\text{rec}}^c = \frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{S}\left\|x_i^s - \tilde{x}_i^{c-s}\right\|_2^2 \tag{7}$$

where x_i^s refers to the original multisensor industrial data, and $\tilde{x}_i^{c-s}$ indicates the reconstruction of the industrial data in Commonality-net.

C. Individuality-Net

The Individuality-net is designed to extract specific representation features and recover the reconstruction of multisensor industrial data. The encoder transforms the multisensor industrial data into specific representation features, while the decoder facilitates the reconstruction data based on specific representation features.

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The first step of Individuality-net is to transform the input multisensor industrial data x_i^s into specific representation features through an encoder layer

$$z_i^s = f_{\text{en}}\left(w_{\text{en}} x_i^s + b_{\text{en}}\right) \tag{8}$$

where z_i^s is the specific representation feature of sth sensor, f_{en} is the activation function of the encoding layer, and w_{en} and b_{en} denote the weight matrix and bias of the encoder layer, respectively, and then the decoder layer converts the specific representation features into the reconstructed industrial data

$$\tilde{x}_i^s = f_{\text{de}}\left(w_{\text{de}} z_i^s + b_{\text{de}}\right) \tag{9}$$

where $\tilde{x}_i^s$ is the reconstruction of the input multisensor industrial data, f_{de} is the activation function of the decoding layer, and w_{de}, b_{de} are the weight matrix and bias of the decoder layer, respectively.

The parameters of the Individuality-net are optimized by minimizing the reconstruction error between x_i^s and $\tilde{x}_i^s$

$$L_{\text{rec}}^I = \frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{S}\left\|x_i^s - \tilde{x}_i^s\right\|_2^2. \tag{10}$$

Up to this point, the complete reconstruction loss can be summarized as follows:

$$L_{\text{rec}} = L_{\text{rec}}^C + L_{\text{rec}}^I = \frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{S}\left\|x_i^s - \tilde{x}_i^{c-s}\right\|_2^2 + \frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{S}\left\|x_i^s - \tilde{x}_i^s\right\|_2^2. \tag{11}$$

In real-world industry, industrial data is generally collected from multiple sensors [50]. Based on this, the FD method using multisensor data has been investigated. For example, Jiang et al. [51] used a contrastive learning framework to achieve cross-domain FD. Similarly, Hu et al. [52] applied cross-sensor self-supervised learning to enhance feature extraction and improve FD in noisy environments.

A consensus-specific constraint is proposed to take consistent and complementary information into account for exploring the consensus representation feature. As for consistent information, a commonality loss is developed based on the extracted features, utilizing Euclidean distance as a means of measuring the distance between the consensus representation feature and the specific representation features. As for complementary information, an individuality loss that is a counterbalance to the commonality loss has been developed, measuring the distance between the specific representation features of different sensors

$$L_C = \sum_{i=1}^{N}\sum_{s=1}^{S}\left\|z_i^s - z_i^c\right\|_2^2 \tag{12}$$

$$L_I = \sum_{i=1}^{N}\sum_{m\neq n}^{S}\left\|z_i^m - z_i^n\right\|_2^2 \tag{13}$$

where z_i^m and z_i^n denote the specific representation features of different sensors.

By shrinking the commonality loss function, the consensus representation feature is optimized. In essence, this loss drives the network to focus on shared characteristics across sensors that are critical for clustering tasks, improving the model's ability to learn a consensus representation feature that captures the global structure of the data. Meanwhile, by expanding the individuality loss, the consensus representation feature is further influenced. In essence, this loss encourages the network to retain complementary information, ensuring that each sensor's unique aspects are still preserved. Together, the consensus-specific constraint ensures that while the model focuses on extracting consensus representation features, it does not lose the distinction of each sensor, which is crucial for FD.

D. Joint Multisensor Deep Clustering Learning Strategy

The proposed method MsDCN involves enhancing the clustering performance by leveraging KL divergence that operates through a two-step iterative process. Initially, an estimate of the nonlinear mapping function and cluster centroids is assumed. The first step consists of computing a soft assignment between the deep representation features (both consensus representation features and specific representation features) and the cluster centroids. In the second step, both the deep mapping function and the cluster centroids are updated based on the current assignments with high confidence using an auxiliary target distribution. This process is repeated iteratively until a convergence criterion is satisfied.

The student's t-distribution serves as the soft assignment distribution in MsDCN for assessing the similarity between the deep representation features and cluster centroids. The soft assignment distribution $q_{i,j}^{c/s}$, which means the probability of the ith instance to the jth cluster, is calculated as

$$q_{i,j}^{c/s} = \frac{\left(1 + \left\|z_i^{c/s} - \mu_j\right\|^2\right)^{-1}}{\sum_j \left(1 + \left\|z_i^{c/s} - \mu_j\right\|^2\right)^{-1}} \tag{14}$$

where $z_i^{c/s}$ denotes the consensus representation features or specific representation features, μ_j denotes jth cluster centroid, and then the auxiliary target distribution $p_{i,j}^{c/s}$ is defined as

$$p_{i,j}^{c/s} = \frac{\left(q_{i,j}^{c/s}\right)^2 \Big/ \sum_i q_{i,j}^{c/s}}{\sum_j \left(\left(q_{i,j}^{c/s}\right)^2 \Big/ \sum_i q_{i,j}^{c/s}\right)} \tag{15}$$

where the target distribution $p_{i,j}^{c/s}$ is defined by boosting the high confidence grades of the soft assignment distribution $q_{i,j}^{c/s}$.

The clustering loss is quantified as the KL divergence between two probability distributions $P_i = [p_{i,j}^c, p_{i,j}^1, p_{i,j}^2, \ldots, p_{i,j}^S]$ and $Q_i = [q_{i,j}^c, q_{i,j}^1, q_{i,j}^2, \ldots, q_{i,j}^S]$. Mathematically speaking, the clustering loss can be expressed as

$$L_{\text{KL}} = \sum_i^N \text{KL}(P_i||Q_i) = \sum_{i=1}^{N}\sum_{j=1}^{K} p_{i,j}^{c/s} \log \frac{p_{i,j}^{c/s}}{q_{i,j}^{c/s}} \tag{16}$$

where KL divergence serves as a means to quantify the nonsymmetric difference between two given probability distributions P_i and Q_i, which are defined in (14) and (15), respectively.

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Based on the reconstruction loss, KL divergence, and consensus-specific constraint mentioned above, this article presents a joint multisensor deep clustering learning strategy that optimizes feature extraction and fault recognition simultaneously, which shapes the objective function

$$\begin{aligned} L_{\text{total}} &= L_{\text{rec}} + \lambda_0 L_{\text{KL}} + \lambda_1 L_C - \lambda_2 L_I \\ &= \frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{S}\left\|x_i^s - \tilde{x}_i^{c-s}\right\|_2^2 + \frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{S}\left\|x_i^s - \tilde{x}_i^s\right\|_2^2 \\ &\quad + \lambda_0 \sum_{i=1}^{N}\sum_{j=1}^{K} p_{i,j}^{c/s}\log\frac{p_{i,j}^{c/s}}{q_{i,j}^{c/s}} \\ &\quad + \lambda_1 \sum_{i=1}^{N}\sum_{s=1}^{S}\left\|z_i^s - z_i^c\right\|_2^2 - \lambda_2 \sum_{i=1}^{N}\sum_{m\neq n}\left\|z_i^m - z_i^n\right\|_2^2 \end{aligned} \tag{17}$$

where L_{rec}, L_{KL}, L_C, and L_I represent reconstruction loss, clustering loss, commonality loss, and individuality loss, respectively. In addition, λ_0, λ_1, and λ_2 denote three coefficients of different loss functions. The primary objective of the joint multisensor deep clustering learning strategy is to enhance the performance of clustering tasks jointly through the sufficient exploration of the consensus representation feature with consistent and complementary information.

E. Optimization Procedure

To address the optimization problem of the network architecture with a separate learning strategy, we propose a joint multisensor learning strategy whereby both deep representation features extraction and clustering are optimized simultaneously. This framework consists of two distinct procedures aimed at achieving optimal performance. The first procedure involves pre-training the multisensor deep networks to obtain optimized parameters of MsDCN. The second procedure is an iterative optimization process that simultaneously optimizes the networks and clustering. Through this strategy, the overall loss functions are minimized to generate the most optimal set of parameters.

1) Pre-Training: The pre-training process involves the optimization of parameters in both the Commonality-net and Individuality-net. Initially, the multisensor deep networks generate a consensus representation feature, specific representation features, and reconstruction data. Subsequently, minimizing the reconstruction loss results in the pre-training of multisensor deep networks and the acquisition of a set of optimized parameters. To update the parameters of AEs from the networks, we utilize a combination of the Adam optimizer and the backpropagation (BP) algorithm.

2) Fine-Tuning: Fine-tuning refers to the formal training process of the proposed MsDCN, in which an iterative optimization process is employed to refine the parameters of the networks and clustering task. First, the parameters obtained from pre-training serve as the initial parameters for the Commonality-net and Individuality-net. Subsequently, the parameters of the Commonality-net with SaAE, Individuality-net, and the initial cluster centroids obtained via the K-means algorithm, and the auxiliary target distribution are updated by minimizing the overall loss functions. Finally, the proposed MsDCN acquires the most optimal parameters. The final clustering results are obtained using the following calculation:

$$c_i = \arg\max_{j=1,2,\ldots,K} q_{i,j}^c. \tag{18}$$

Table I illustrates the pseudocode of the proposed method.

TABLE I
PSEUDOCODE OF THE PROPOSED MSDCN

Input: Dataset $D = \{x_i^1, x_i^2, ..., x_i^S\}_{i=1}^N$, where $x_i^s \in \mathbb{R}^{d_s}$;
Number of clusters K;
Coefficient of clustering loss λ_0;
Coefficient of commonality loss λ_1;
Coefficient of individuality loss λ_2;
1. **Pre-training phase:**
1.1. Apply the SaAE to assign attention weights:
1.1.1. Compute GAP and GMP features using equations (1) and (2)
1.1.2. Concatenate the extracted features and pass them through an MLP.
1.1.3. Obtain sensor attention weights w_i^c using equation (3).
1.1.4. Apply element-wise multiplication to compute sensor-aware features x_i^c using equation (4).
1.2. Train the Commonality-net and Individuality-net by minimizing the reconstruction loss of each sensor using equations (7) and (10).
1.3. Calculate consensus representation feature z_i^c and specific representation features z_i^s according to Section III-B and III-C.
1.4. Initialize cluster centroids μ_j by the K-means algorithm.
2. **Fine-tuning phase:**
2.1. **while** not reaching the maximum iterations **do:**
2.1.1. Compute sensor-aware features x_i^c via the SaAE as in Step 1.
2.1.2. Calculate consensus representation feature z_i^c and specific representation features z_i^s according to Section III-B and III-C.
2.1.3. Compute the soft assignment distribution: $Q_i = [q_{i,j}^c, q_{i,j}^1, q_{i,j}^2, ..., q_{i,j}^S]$ by equation (14).
2.1.4. Compute the auxiliary target distribution: $P_i = [p_{i,j}^c, p_{i,j}^1, p_{i,j}^2, ..., p_{i,j}^S]$ by equation (15).
2.1.5. Update the parameters of the framework by minimizing the total loss function described in equation (17).
2.2. **end while**
3. **Output:**
Multi-sensor cluster assignment c_i calculated by equation (18).

IV. EXPERIMENT SETUP

A series of experiments were conducted to assess the efficacy of the proposed MsDCN, and the experimental results are presented in this article. Initially, a comprehensive description of the datasets employed in these experiments is provided. Subsequently, a set of appropriate evaluation metrics is selected to accurately assess the performance of the MsDCN. Finally, the experimental procedure is introduced in detail.

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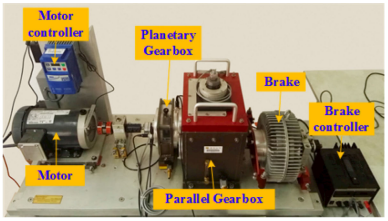


Fig. 3. Structure of the DDS.

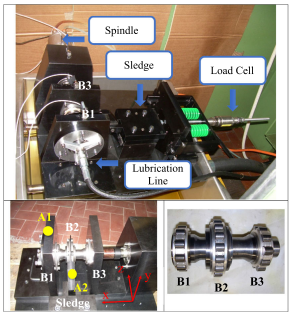


Fig. 4. Structure of the test rig.

A. Datasets

This study utilizes two datasets for evaluation, i.e., one from Southeast University (SEU) and another from the Politecnico di Torino, provided by the Dynamic and Identification Research Group (DIRG). The SEU dataset was obtained through data collection from the drivetrain dynamic simulator (DDS), as depicted in Fig. 3 [53]. It comprises gearbox and bearing data, each encompassing distinct operational conditions with rotational speed-load configurations (20 Hz–0 V and 30 Hz–2 V). Vibration signals are collected using eight sensors, i.e., three mounted on the planetary and parallel gearboxes along the x-, y-, and z-axes to capture vibrations in all directions, and two additional sensors mounted on the motor for vibration and torque detection. As shown in Fig. 4, the DIRG dataset was acquired from the test rig [54]. It includes vibration signals recorded from six sensors on the bearing, comprising six fault states and one healthy state. The six sensors capture accelerations in different directions, i.e., two measure axial vibrations along the x-axis, while the remaining four record radial vibrations along the y- and z-axes. By incorporating data from an additional dataset with varying fault characteristics, MsDCN's capacity to generalize across diverse industrial scenarios is further examined. Details of these datasets are shown in Tables II and III.

As detailed in Table II, the SEU dataset comprises four sub-datasets, each containing distinct working conditions. Two of these sub-datasets pertain to gearboxes and encompass healthy state (H), surface fault (SuF), root fault (RF), missing tooth (MT), and chipped tooth (CT). The remaining two sub-datasets relate to bearings and include healthy state (H), outer ring fault (ORF), inner ring fault (IRF), combination tooth fault (CTF), and ball fault (BF). Each instance in the SEU dataset involves eight sensors, capturing vibration signals along different axes, with a data length of 2048 per sensor. Additionally, the DIRG rolling bearing dataset consists of vibration signals recorded from six sensors under seven working states, including one healthy state and six fault types. The fault types include inner ring indentation (450/250/150 μm) and roller indentation (450/250/150 μm) with varying levels of severity. Further insights into the occurrence locations and fault conditions are presented in Table III.

TABLE II
DETAILS OF THE SIX DISTINCT DATASETS OF GEARBOX AND BEARING

Location	Datasets	Number of clusters	Instance sets	working conditions	Working conditions
Gearbox	D1	5	5112*8*2048	H, SUF, RF, MT, CT	30HZ-2V
	D2	5			20HZ-0V
Bearing	D3	5	5112*8*2048	H, ORF, IRF, CTF, BF	30HZ-2V
	D4	5			20HZ-0V
	D5	7	700*6*5120	0A, 1A, 2A, 3A, 4A, 5A, 6A	300HZ-1KN
	D6	7			100HZ-1.8KN

TABLE III
FAULT TYPES DESCRIPTION OF GEARBOX AND BEARING

Location	Fault types	Description of the fault types
Gearbox	Surface fault	Wear occurs on the surface of the gear
	Root fault	Crack occurs in the root of the gear feet
	Missing tooth	One of the gear teeth is missing
	Chipped tooth	Crack occurs in the gear tooth
Bearing	Outer ring fault	Crack occurs in the outer ring
	Inner ring fault	Crack occurs in the inner ring
	Combination fault	Crack occurs in the inner and outer ring
	Ball fault	Crack occurs in the ball
	1A, 2A, 3A	Inner ring indentation (450/250 /150μm)
	4A, 5A, 6A	Roller indentation (450/250/150μm)

B. Evaluation Metrics

To assess the clustering performance of the proposed MsDCN and other benchmark methods, three widely accepted clustering evaluation metrics have been employed in this article. Specifically, the clustering accuracy (ACC), the normalized mutual information (NMI), and the adjusted rand index (ARI) have been utilized as evaluation metrics [55], [56].

In specific, the ACC aims to measure the accuracy of the best mapping between the cluster assignments and the true labels, and the best mapping is computed by using the Hungarian algorithm. The ACC can be formulated as

$$\mathrm{ACC} = \max_{\mathrm{map}} \frac{\sum_{i=1}^{N} I\{y_i = \mathrm{map}(c_i)\}}{N} \tag{19}$$

where y_i represents the true label of ith instance, c_i represents the cluster assignment of ith instance, and $\mathrm{map}(c_i)$ denotes the possible one-to-one mapping between the real labels and the predictive results.

The NMI represents the normalization measure of similarity that is utilized to evaluate the degree of correspondence between the predicted labels and the true labels. The calculation equation of the NMI is defined as follows:

$$\mathrm{NMI} = \frac{I(y_i; c_i)}{\max\{H(y_i), H(c_i)\}} \tag{20}$$

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TABLE IV
DETAILS OF THE PARAMETER SETTINGS IN THE EXPERIMENTS

Methods	parameters
K-means	Initialization: K-means++; Distance Metric: Euclidian distance.
AE+K-means	Encoder network dimensions: 16384-250-250-500-10; Decoder network dimensions: 10-500-250-250-16384.
AE+DenMune [57]	Encoder network dimensions: 16384-250-250-500-10; Decoder network dimensions: 10-500-250-250-16384.
AE+HIAC [58]	Encoder network dimensions: 16384-250-250-500-10; Decoder network dimensions: 10-500-250-250-16384.
DEC [47]	Encoder network dimensions: 16384-250-250-500-10; Decoder network dimensions: 10-500-250-250-16384.
IDEC [48]	Encoder network dimensions: 16384-250-250-500-10; Decoder network dimensions: 10-500-250-250-16384; Coefficient of clustering loss: 0.5.
DCN [44]	Encoder network dimensions: 16384-256-256-512-10; Decoder network dimensions: 10-512-256-256-16384; Coefficient of clustering loss: 0.3.
DKM [45]	Encoder network dimensions: 16384-256-256-512-10; Decoder network dimensions: 10-512-256-256-16384; Coefficient of clustering loss: 0.3.
CVCL [59]	Encoder network dimensions: 16384-250-250-500-10; Decoder network dimensions: 10-500-250-250-16384; Coefficient of cross-view contrastive loss: 0.001; Coefficient of cross-view consistency loss: 0.001.
EM-AAE [60]	Encoder network dimensions: 16384-250-250-500-80; Decoder network dimensions: 80-500-250-250-16384; Discriminator network dimensions: 80-128-256-1.
SensorSCAN [61]	Encoder network dimensions: Conv1d(8,32,3)-Conv1d(32,64,3)-Linear(65536, 80); Decoder network dimensions: Linear(80, 65536)-Conv1d(64,32,3)-Conv1d(32,8,3); Coefficient of contrastive loss: 0.5.
MsDCN	Encoder network dimensions: 2048-256-256-512-d; Decoder network dimensions: d-512-256-256-2048; Dimension of consensus representation feature and specific representation features d: {10, 20, 40, 80, 160}; Coefficient of clustering loss: {0.1, 1, 10, 100, 500, 1000}; Coefficient of individuality loss: {0.0001, 0.001, 0.01, 0.1, 1, 10}; Coefficient of commonality loss: {0.0001, 0.001, 0.01, 0.1, 1, 10}.

where $I(y_i; c_i)$ denotes the mutual information between the true label y_i and the cluster assignment c_i. $H(y_i)$ represents the entropy of the true label y_i, and $H(c_i)$ represents the entropy of the cluster assignment c_i. Importantly, it should be noted that the NMI metric is invariant to the permutations of the clusters.

The ARI serves as an enhanced version of the rand index (RI). In specific, it is utilized to evaluate the similarity between two given clusters by considering all pairs of instances and counting the pairs that are assigned in the same or different clusters in the predicted clusters and ground truth. The ARI is formulated as follows:

$$\mathrm{ARI} = \frac{\mathrm{RI} - E(\mathrm{RI})}{\max(\mathrm{RI}) - E(\mathrm{RI})} \tag{21}$$

where $E(\mathrm{RI})$ represents the expectation of the RI, and $\max(\mathrm{RI})$ expresses the maximum value of RI.

$F1$ score is a common supervised learning performance metric. To measure the comprehensiveness of the model performance, we compute the $F1$ score on different datasets by aligning predicted cluster labels with true labels. The form is defined as follows:

$$F1\ \text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

where Precision and Recall are computed based on true positive (TP), false positive (FP), and false negative (FN) obtained from optimal cluster-label assignments determined by the Hungarian algorithm.

C. Experimental Procedure

To demonstrate the superiority of the proposed MsDCN, we conducted a comparative analysis between the proposed method and various benchmark clustering methodologies, namely K-means, AE + K-means, AE + DenMune [57], AE + HIAC [58], DEC [47], IDEC [48], DCN [44], DKM [45], CVCL [59], EM-AAE [60], and SensorSCAN [61].

The K-means is representative of traditional machine learning approaches, while others comprise deep clustering techniques. Among them, AE + K-means, AE + DenMune, and AE + HIAC entail a two-stage process with a separate learning strategy. DEC, IDEC, DCN, DKM, and CVCL are deep clustering methods with a joint learning strategy for optimization, while EM-AAE and SensorSCAN are state-of-the-art deep clustering-based FD methods. The specific parameters for these methods can be found in Table IV.

To reduce the potentially detrimental impact of randomness, the experimental procedures for each method were iteratively performed 20 times. Then, the average values of every evaluation metric were calculated to assess the final clustering performance, thereby arriving at a reliable and robust estimate of the overall clustering performance.

V. RESULTS AND DISCUSSION

This section presents a comprehensive analysis and comparison of the experimental results obtained through the proposed MsDCN and other benchmark clustering methods. The evaluations of various aspects of the MsDCN are meticulously scrutinized and discussed in detail.

A. Results

The experimental results of the proposed method and comparison methods are compared and analyzed. Table V shows the mean and standard deviation values associated with three evaluation metrics across different datasets and methods,

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TABLE V
COMPARISON OF CLUSTERING PERFORMANCE IN TERMS OF ACC (%), NMI (%), ARI (%), AND *F*1 SCORE (%) ON D1 AND D2

	D1				D2			
Method	ACC	NMI	ARI	F1 score	ACC	NMI	ARI	F1 score
K-means	79.58±0.06***	87.64±0.07***	77.03±0.12***	74.10±0.03***	79.41±0.04***	84.44±0.03***	75.69±0.02***	74.82±0.01***
AE+K-means	96.88±6.02***	95.55±3.22***	95.01±6.51***	97.60±5.58***	96.52±5.80***	94.80±2.98***	94.10±6.08***	93.74±9.52***
AE+DenMune	97.57±2.93***	96.08±1.40**	96.41±2.04**	88.87±14.71**	97.51±4.33***	96.86±1.74**	96.35±4.40**	88.39±1.70**
AE+HIAC	98.32±0.44***	95.23±1.18***	95.83±1.10***	83.82±11.91***	96.76±1.16*	92.21±2.44*	92.23±2.59*	81.62±1.83*
DEC	98.22±4.44***	97.04±2.49**	96.95±4.88***	97.61±5.57***	98.77±0.59***	96.61±1.59***	96.96±1.46***	98.47±0.40***
IDEC	99.08±0.29***	97.34±0.52***	97.74±0.70***	96.54±1.68***	99.01±0.50	97.34±1.21	97.58±1.22	99.04±0.51
DCN	99.26±0.34***	97.63±1.00***	98.17±0.85***	97.44±1.08***	98.97±0.39***	97.14±0.98***	97.48±0.95***	98.85±0.38***
DKM	99.16±0.27**	97.43±0.77**	97.91±0.68**	97.88±0.60**	98.88±0.45***	96.94±1.14***	97.24±1.10***	98.74±0.46***
CVCL	99.12±0.25***	97.27±0.68***	97.84±0.60***	99.08±0.42***	99.13±0.17*	97.39±0.43*	97.83±0.41*	99.13±0.94*
EM-AAE	99.10±0.20***	97.31±0.60***	97.75±0.50***	98.78±0.39***	98.16±0.28***	95.41±0.73***	95.48±0.70***	98.10±0.15***
SensorSCAN	99.00±0.18***	97.06±0.49***	97.50±0.44***	91.92±0.34***	98.47±0.10***	95.95±0.22***	96.22±0.26***	96.51±0.11***
MsDCN	**99.52±0.19**	**98.42±0.57**	**98.82±0.47**	**99.58±0.18**	**99.24±0.38**	**98.04±0.90**	**98.14±0.92**	**99.24±0.38**

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. The best results are in bold

TABLE VI
COMPARISON OF CLUSTERING PERFORMANCE IN TERMS OF ACC (%), NMI (%), ARI (%), AND *F*1 SCORE (%) ON D3 AND D4

	D3				D4			
Method	ACC	NMI	ARI	F1 score	ACC	NMI	ARI	F1 score
K-means	78.44±0.03***	88.00±0.03***	76.29±0.03***	78.42±0.02***	76.32±0.05***	83.12±0.11***	72.31±0.13***	76.32±0.05***
AE+K-means	96.63±5.73***	95.42±2.57**	94.44±5.78***	96.46±0.09***	96.70±5.85***	94.85±3.38***	94.51±6.34***	96.21±7.16***
AE+DenMune	95.40±7.28***	94.67±3.53**	93.08±7.80**	94.30±8.95***	97.53±0.90***	95.01±1.43***	95.37±1.61***	92.45±1.21***
AE+HIAC	96.63±5.73***	95.42±2.57**	94.44±5.78***	95.21±8.14**	97.24±1.19**	93.85±2.42**	93.32±2.88**	85.24±3.59**
DEC	98.60±0.10***	96.55±0.31**	96.59±0.25***	98.42±0.09**	98.20±4.32**	96.92±2.64**	96.90±4.81**	96.32±7.17**
IDEC	98.76±0.07*	96.99±0.20	96.98±0.17*	98.71±0.10*	99.08±0.29*	97.34±0.52**	97.74±0.70*	97.10±0.29**
DCN	98.72±0.10***	96.88±0.25***	96.90±0.23***	98.60±0.08***	99.04±0.37***	97.31±0.61***	97.66±0.87***	98.76±0.31***
DKM	98.70±0.07*	96.84±0.21	96.84±0.17*	98.64±0.17*	98.85±0.39***	97.01±0.64***	97.22±0.93***	99.06±0.34***
CVCL	98.72±4.36*	96.55±2.80**	97.14±5.00*	96.90±0.60*	98.77±0.40**	96.30±1.09**	96.99±0.96**	99.03±0.38**
EM-AAE	98.75±0.07*	96.95±0.29	96.95±0.18*	98.64±0.12*	98.90±0.30***	97.00±0.68***	97.46±0.72***	98.91±0.40***
SensorSCAN	98.70±0.07**	96.84±0.21*	96.84±0.17**	98.74±0.12**	98.04±0.37***	95.06±0.75***	95.18±0.90***	98.16±0.40***
MsDCN	**98.94±0.38**	**97.12±0.94**	**97.42±0.92**	**98.82±0.47**	**99.32±0.30**	**97.88±0.79**	**98.32±0.72**	**99.42±0.29**

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. The best results are in bold

TABLE VII
COMPARISON OF CLUSTERING PERFORMANCE IN TERMS OF ACC (%), NMI (%), ARI (%), AND *F*1 SCORE (%) ON D5 AND D6

	D5				D6			
Method	ACC	NMI	ARI	F1 score	ACC	NMI	ARI	F1 score
K-means	26.14±3.49***	12.28±3.93***	5.91±3.07***	20.79±3.47***	23.91±1.41**	38.23±4.77***	17.75±3.69***	18.26±4.72***
AE+K-means	28.84±4.38***	31.62±4.35*	11.79±2.61**	27.54±3.00*	31.47±2.50**	42.25±4.56**	21.06±3.10**	27.05±3.36**
AE+DenMune	27.75±8.21***	26.56±2.53***	7.35±1.81***	20.37±9.35***	30.52±3.24**	28.13±5.32**	19.49±3.95**	27.38±5.71**
AE+HIAC	25.59±4.98**	17.75±8.80*	6.49±2.91*	21.48±6.71**	30.56±1.61**	29.15±4.63**	17.34±4.71**	21.63±5.27**
DEC	28.47±3.74**	30.97±3.28*	12.06±2.41*	29.48±3.01**	41.33±3.01**	49.48±4.53**	28.13±3.47**	31.46±3.72**
IDEC	30.28±2.93**	35.03±4.75	13.68±3.46**	19.51±5.82**	40.68±1.32**	53.48±3.69**	33.79±4.68**	36.18±4.83
DCN	31.20±4.96**	33.41±3.94*	14.94±3.69*	28.57±3.92*	39.60±1.53**	53.72±3.43**	22.34±3.50**	31.43±3.82**
DKM	31.54±2.95**	37.37±2.88***	16.78±3.23*	28.21±2.15**	38.71±1.63**	39.22±3.53**	24.34±3.72**	24.34±3.87**
CVCL	37.73±4.12*	50.66±6.39**	31.08±4.35*	31.56±4.71*	41.73±2.11**	54.66±6.11**	25.35±4.05**	31.24±5.73**
EM-AAE	38.01±6.37**	42.90±13.14*	22.96±6.05***	31.56±7.23*	44.59±3.22*	60.99±7.39*	38.61±5.15*	32.73±6.91*
SensorSCAN	40.15±6.36**	60.23±6.31***	28.89±6.57**	35.14±7.91**	46.58±7.08**	55.99±7.39**	33.61±6.68**	21.56±7.90**
MsDCN	**51.50±6.71**	**61.00±5.39**	**38.61±5.67**	**46.89±6.51**	**52.32±6.30**	**62.88±6.79**	**39.32±6.72**	**40.82±6.47**

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. The best results are in bold

where the best results are marked in bold. Further, the number of parameters and computational complexity are displayed in Table VI, which helps to understand the performance of different methods by comparing the properties of these models.

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TABLE VIII
NUMBER OF PARAMETERS, COMPUTATIONAL COMPLEXITY, AND TRAINING TIME OF DIFFERENT CLUSTERING METHODS

Method	The number of model parameters	Computational complexity	Training time (s)
AE+K-means	8.60×10^6	8.60×10^6	78.67
AE+DenMune	8.60×10^6	8.60×10^6	88.36
AE+HIAC	8.60×10^6	8.60×10^6	57.29
DEC	8.61×10^6	8.61×10^6	86.53
IDEC	8.61×10^6	8.61×10^6	86.14
DCN	8.81×10^6	8.81×10^6	139.78
DKM	8.81×10^6	8.81×10^6	175.87
CVCL	1.13×10^7	1.13×10^7	271.48
EM-AAE	1.19×10^7	1.19×10^7	187.08
SensorSCAN	1.06×10^7	7.28×10^7	211.43
MsDCN	2.64×10^7	2.66×10^7	207.31

As shown in Tables V–VII, first of all, the majority of the results of MsDCN are statistically significant from the Wilcoxon matched-pairs signed rank test of the comparison methods. The MsDCN obtains outstanding clustering performance in all six datasets, showing the effectiveness of the proposed method. Second, the deep clustering methods obtain more advanced clustering performance than the K-means method under the above evaluation metrics, verifying the superiority of the deep clustering methods compared with traditional clustering methods. Finally, the proposed MsDCN achieves lower standard deviation values than other benchmark methods in most cases, indicating strong steadiness.

Furthermore, we analyzed the number of model parameters, computational complexity, and training time. As shown in Table VIII, first, traditional two-stage methods like AE + K-means, AE + DenMune, and AE + HIAC demonstrate similar model parameter numbers and computational complexities, approximately at the 10^6 scale, and their training times are relatively moderate, generally within 90 s. Conversely, other methods enhance FD performance by utilizing more parameters, computational complexity, and longer training times. For example, CVCL and EM-AAE methods require significantly higher training time, over 270 and 187 s, respectively, suggesting that an appropriate increase in model size and complexity can enhance FD capabilities. Second, the proposed MsDCN, although having the number of model parameters and computational complexity on the order of 10^7, achieves comparatively moderate training time (207.31 s), indicating that MsDCN effectively balances computational demands with diagnostic performance, outperforming methods like CVCL and EM-AAE in terms of efficiency. Third, the moderate training time and complexity of MsDCN suggest its viability for real-world industrial scenarios from a practical perspective. Industrial FD requires rapid training due to evolving data patterns. Since MsDCN achieves superior clustering performance (see Tables V–VII) with acceptable computational cost, it effectively balances accuracy and efficiency, making it suitable for practical deployment.

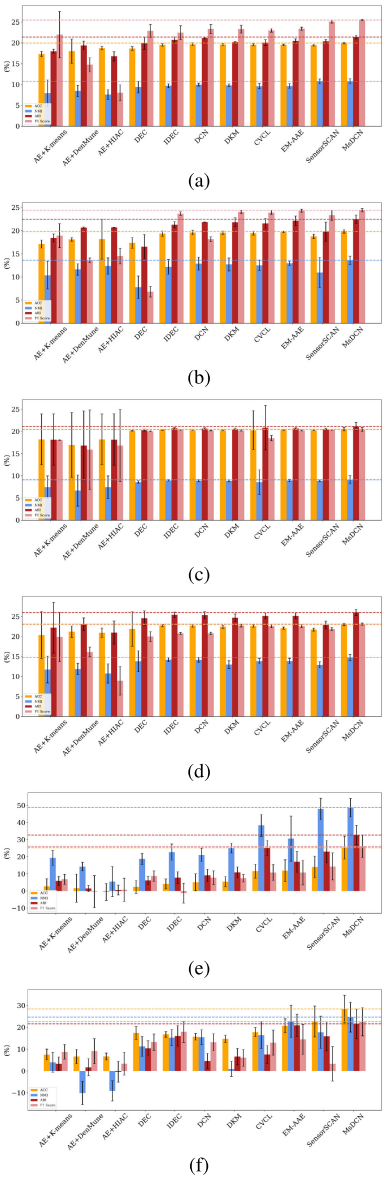


Fig. 5. Comparisons of improved performance of different deep clustering methods on six datasets. (a) D1. (b) D2. (c) D3. (d) D4. (e) D5. (f) D6.

B. Discussion

1) Evaluation of Different Clustering Methods: To validate the effectiveness of the proposed MsDCN, we investigated the improved diagnostic performance of different deep clustering methods on six datasets. The improved evaluation metrics of AE + K-means, AE + DenMune, AE + HIAC, DEC, IDEC, DCN, DKM, CVCL, EM-AAE, SensorSCAN, and the proposed MsDCN compared with K-means are shown in Fig. 5.

As shown in Fig. 5, first, the experimental results of all the deep clustering methods are improved as compared to the traditional methods. For example, the ACC values of the proposed MsDCN under six datasets are improved by 19.94% (D1), 19.83% (D2), 20.60% (D3), 23.00% (D4), 25.36% (D5), and 28.41% (D6) in comparison with K-means. The results of the comparative analysis reveal that the FD performance of deep clustering methods outperforms the traditional clustering methods. Second, the comparative results of deep clustering

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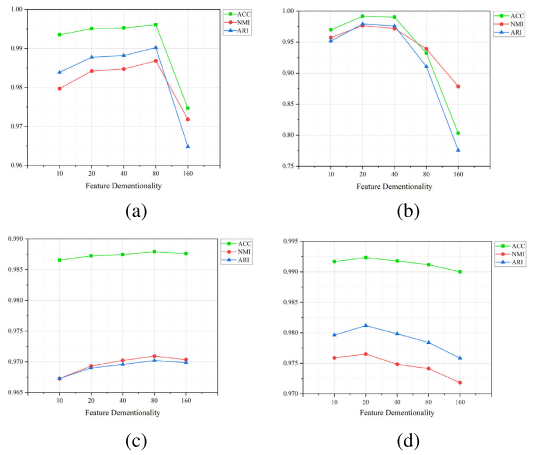


Fig. 6. Comparisons of different dimensionalities of consensus representation feature on SEU datasets. (a) D1. (b) D2. (c) D3. (d) D4.

methods with a separate learning strategy and other deep clustering methods with a joint learning strategy, especially the proposed MsDCN, show the superiority in FD. In specific, using ACC values as an example, this evaluation metric of the proposed MsDCN has improved by 2.64% (D1), 2.72% (D2), 2.41% (D3), 2.62% (D4), 22.66% (D5), and 20.85% (D6) in comparison with AE + K-means, respectively. These results support the advantages of the joint learning strategy over the separate learning strategy in extracting clustering-friendly features. Finally, the proposed MsDCN achieved the best clustering performance among joint deep clustering methods and deep clustering-based FD methods. Existing methods struggle in multisensor FD due to their inability to model cross-sensor correlations and shared patterns. For example, deep clustering-based FD methods process each sensor independently, failing to utilize complementary and consensus information, which results in suboptimal clustering performance. In contrast, MsDCN effectively overcomes these limitations. The Commonality-net extracts the consensus representation feature to capture critical global information, while the Individuality-net retains sensor-specific representation features to preserve local patterns. The joint learning strategy optimizes the consensus representation feature, ensuring it is both clustering-friendly and discriminative, enabling MsDCN to balance complementary and consistent information, thereby enhancing FD performance in multisensor scenarios.

2) Evaluation of Different Dimensionalities of Consensus Representation Feature: The dimensionality of the consensus representation feature plays a crucial role in FD. Therefore, we discuss the influence of varying dimensionalities of the consensus representation feature on clustering performance. The detailed depiction is shown in Fig. 6.

As shown in Fig. 6, first of all, the optimal clustering performance is obtained when the dimensionality of the consensus representation feature is set to 80 (D1), 20 (D2), 80 (D3), and 20 (D4), which validates that distinct dimensionality of consensus representation feature can maximize the FD performance in different datasets. Second, upon examining the dimensionality range from 10 to 160, it is observed that the evaluation metrics, including the ACC, NMI, and ARI, initially exhibit an upward trend, while these metrics gradually decline beyond an optimal point. Considering dataset D1 as an example, the evaluation metrics exhibit a gradual increase as the dimensionality is varied from 10 to 80, and the optimal performance is achieved when the dimensionality is set to 80. Notably, the most significant improvement is observed in the range of 10–20 dimensions, and when the dimensionality is increased from 80 to 160, there is a significant decrease in the evaluation metrics. This phenomenon can be attributed to the fact that when the dimensionality of the consensus representation feature is too small, the consensus representation feature fails to encompass an adequate representation of the industrial characteristics, and progressively higher dimensionality of the consensus representation feature holds more critical information. The excessively large dimensionality can cause the extracted feature to include redundant non-critical information, leading to poor clustering performance by adversely affecting the training of the proposed MsDCN.

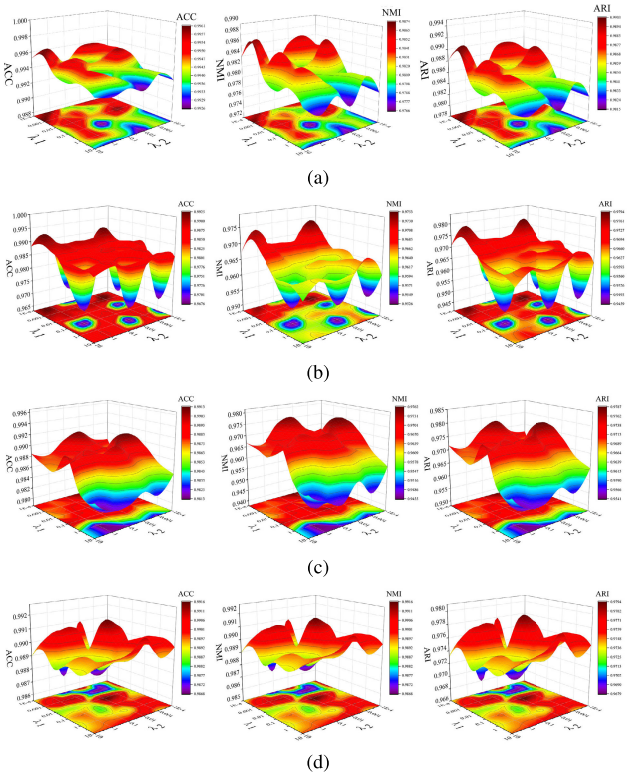


Fig. 7. Sensitive analysis of the evaluation metrics of the proposed MsDCN on SEU datasets. (a) D1. (b) D2. (c) D3. (d) D4.

3) Sensitive Analysis of the Proposed MsDCN: The clustering performance of the MsDCN is significantly influenced by the coefficients of two key loss functions in the proposed consensus-specific constraint: the coefficient λ_1 of the commonality loss and the coefficient λ_2 of the individuality loss. Fig. 7 provides a detailed illustration of the impact of these coefficients on the performance of MsDCN.

As shown in Fig. 7, the proposed MsDCN achieves the best ACC with $\lambda_1 = 0.001/\lambda_2 = 10$ (D1), $\lambda_1 = 0.001/\lambda_2 = 0.001$ (D2), $\lambda_1 = 0.0001/\lambda_2 = 0.01$ (D3), and $\lambda_1 = 0.01/\lambda_2 = 0.01$ (D4), and the results of NMI, and ARI are basically consistent with ACC. First, from the perspective of λ_1, the evaluation metrics increase to reach a clear optimum and decline gradually with an increase of λ_1 from 0.0001 to 10. Taking the ACC values of D1 as an example, when λ_2 is

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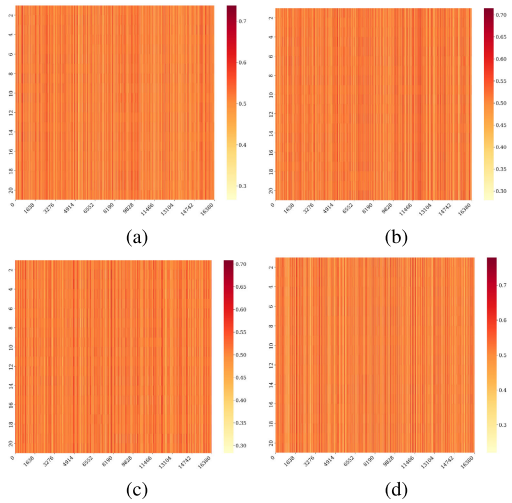


Fig. 8. Attention weights of 20 instances processed by the SaAE on SEU datasets. (a) D1. (b) D2. (c) D3. (d) D4.

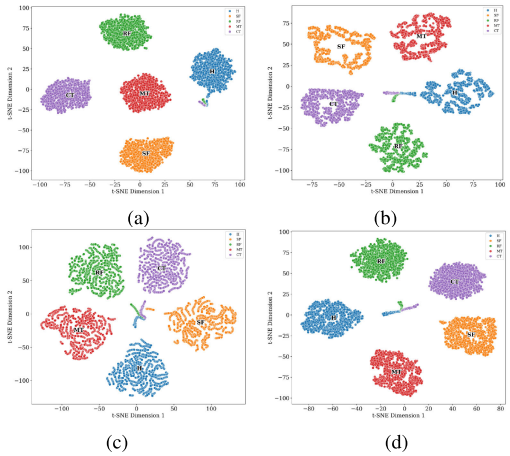


Fig. 9. Visualizations of clustering results of different methods on D1. (a) CVCL. (b) EM-AAE. (c) SensorSCAN. (d) MsDCN.

held constant at 10, and λ_1 is varied from 0.0001 to 10, the overall trend of ACC is first increasing and then gradually declining, despite fluctuations in the decline. This phenomenon can be attributed to that the appropriate utilization of consistent information can positively impact the clustering performance. However, an excessive focus on consistent information leads to redundant components that negatively affect the clustering performance. When λ_1 is set to 10, the mean ACC of MsDCN is not optimal under each dataset; moreover, it is much lower than the best ACC on D3, indicating that the redundant component in D3 is relatively the most numerous. Second, from the perspective of λ_2, the trend of evaluation metrics is not obvious and is accompanied by a certain degree of fluctuation, and the optimal value is not clear with an increase of λ_2 from 0.0001 to 10. Taking the NMI values of D1 for example, when λ_1 is held constant at 0.001, and λ_2 is varied from 0.0001 to 10, a notable pattern emerges. Initially, the ACC values show an upward trend, followed by a gradual decline, and subsequently, another similar up-and-down trend is observed. This phenomenon indicates that complementary information affects clustering performance slightly, and this effect may exist in periodicity. Besides, once λ_1 has been fixed, the effect of λ_2 on the evaluation metrics is more limited, especially when the rotational speed-load configuration is set to 30 Hz–2 V (D1, D3). That is to say, although the complementary information affects the clustering performance by influencing the optimization of the consensus representation feature, consistent information is still the main influencing factor, and the importance of consistent information will be more distinctive under specific working conditions.

4) Analysis of Sensor-Aware Attentive Encoder: The MsDCN incorporates a SaAE, specifically designed to assign varying weights to industrial data based on the relative importance of multiple sensors. In this experiment, 20 instances are obtained via stratified sampling across different working conditions, and their attention weights are visualized in Fig. 8.

As shown in Fig. 8, darker colors indicate higher attention weights, while brighter colors represent lower weights. First, across a single instance, the attention weights of different data points vary significantly, indicating that the model prioritizes certain time steps in multisensor data. Second, within the same sensor acquisition range but across different instances, similar color distributions are often observed, implying that data from the same sensor tends to retain consistent importance across different instances. To quantitatively analyze this phenomenon, the Shannon entropy is calculated across instances and sensors. Take D3 as an example, the cross-instance entropy analysis reveals that the maximum entropy is 5.3193 (instance 17), while the minimum is 4.8874 (instance 19), demonstrating that different instances exhibit varying attention distribution patterns. The cross-sensor entropy analysis further shows that the highest entropy was 5.1303 (sensor 6) and the lowest was 4.9633 (sensor 5) in D3, suggesting that certain sensors are prioritized over others. These findings confirm that SaAE dynamically adjusts attention weights based on both instance-specific and sensor-specific importance, allowing the model to balance local adaptability and global generalization, which is critical for multisensor FD.

5) Visualization of Clustering Results: To gain a more intuitive understanding of the FD performance of the proposed MsDCN, we compare it with CVCL, EM-AAE, and SensorSCAN, using t-SNE visualizations on dataset D1. All 5110 sample points from the dataset are used for these comparisons. The results, presented in Fig. 9, highlight the differences in clustering performance among these methods. The confusion matrix in Fig. 10 compares the matching results of predicted and true labels on the SEU datasets.

As shown in Fig. 9, first, MsDCN [see Fig. 9(d)] forms compact, well-separated clusters, ensuring minimal overlap between fault categories and maintaining clear decision boundaries, which indicates its effective joint optimization of deep representation feature extraction and clustering. Second, CVCL [see Fig. 9(a)] also forms distinct clusters but shows more overlap between RF and CT, and some clusters appear less compact, suggesting suboptimal clustering separation. Third, EM-AAE [see Fig. 9(b)] and SensorSCAN [see Fig. 9(c)] exhibit scattered clusters with weak boundaries, making fault separation less reliable. Their feature spaces lack structural coherence, reflecting the limitation of the separate learning strategy, where feature extraction and clustering are separately optimized, leading to suboptimal clustering alignment. Additionally, Fig. 10 further validates the performance

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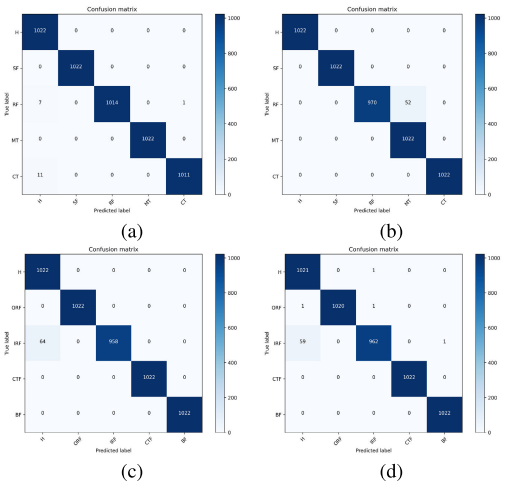


Fig. 10. Confusion matrix of different working conditions on SEU datasets. (a) D1. (b) D2. (c) D3. (d) D4.

TABLE IX
RESULTS OF ABLATION EXPERIMENTS IN TERMS OF ACC (%) ON SEU DATASETS

Methods	D1	D2	D3	D4
– i-loss	99.47±0.26	99.17±0.33	98.63±0.42**	99.08±0.35**
– c-loss	99.32±0.20***	99.02±0.25*	98.65±0.10**	90.10±0.33**
– (i-loss+c-loss)	99.31±0.24*	98.95±0.56**	98.67±0.38**	99.05±0.36**
– SaAE	99.27±0.14**	99.09±0.35*	98.61±0.28**	99.00±0.28***
MsDCN	**99.52±0.19**	**99.24±0.38**	**99.05±0.48**	**99.32±0.30**

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. The best results are in bold

of the proposed method with confusion matrices across SEU datasets. MsDCN achieves high diagnostic accuracy, though minor misclassifications occur, particularly RF in D1 and D2, and IRF in D3 and D4. Despite these, MsDCN consistently outperforms other methods, achieving strong clustering coherence and fault separability.

6) Ablation Experiments: In this section, ablation experiments including structure ablation and loss ablation are conducted to evaluate the contribution of SaAE and learning strategy in the proposed MsDCN. Specifically, structure ablation involves MsDCN without SaAE (− SaAE), while loss ablation sets three different loss ablation groups for joint multisensor deep clustering learning strategy, namely MsDCN without individuality loss (− i-loss), MsDCN without commonality loss (− c-loss), and MsDCN without joint multisensor deep clustering learning strategy (− i-loss and c-loss).

As shown in Table IX, first, the Wilcoxon matched-pairs signed rank test is applied for statistical significance to the experimental results, which indicates that in most cases the ablation experiments exhibit significant or highly significant differences compared to the full proposed method. The comparison of the ACC values of the SEU datasets clearly shows that removing any component from the proposed MsDCN can lead to the degradation of the overall clustering performance, which proves the effectiveness and importance of the SaAE and learning strategies of the proposed MsDCN. Second, after removing SaAE, not only is there a statistically significant difference compared to the results of the full model, but the worst clustering performance is found on D1, D3, and D4, which are 99.27% (D1), 98.61% (D3), and 99.00% (D4), respectively. The observed phenomenon can be attributed to the effectiveness of the Commonality-net in extracting a high-quality consensus representation feature that considers sufficient information, thus playing a pivotal role in enhancing the overall clustering performance.

VI. CONCLUSION

This article proposes an MsDCN with a joint learning strategy for FD, leveraging complementary and consistent information from multisensor data. The framework integrates a Commonality-net, an Individuality-net, and a joint clustering strategy.

Specifically, the Commonality-net with SaAE extracts consensus representation features and reconstructs input data, while the Individuality-net captures specific representation features. Together, these networks optimize feature extraction and clustering through a consensus-specific constraint. The MsDCN addresses two key gaps in existing deep clustering-based FD methods, i.e., the separate learning strategy commonly used leads to suboptimal FD results, and the consensus representation feature in multisensor data is not fully explored, limiting the overall effectiveness of FD. In addition to addressing theoretical challenges, MsDCN also solves practical issues in FD. It handles unlabeled data effectively through its unsupervised learning approach, which makes it highly applicable in environments where labeled data is scarce. The model efficiently processes multisensor data by leveraging both consistent and complementary information, ensuring comprehensive FD across multisensor inputs. We validated the performance of MsDCN with experiments on real-world industrial datasets, and the results demonstrate that the proposed model significantly outperforms other popular deep clustering methods in accurately identifying different working conditions.

Despite the impressive performance of the proposed method, several limitations remain. First, MsDCN lacks explicit noise-robust mechanisms, making it susceptible to performance degradation due to sensor faults and electromagnetic interference that distort feature extraction and clustering quality in industrial data. Second, the method is primarily designed for single-FD, limiting its ability to classify multiple coexisting fault types, which often exhibit overlapping feature patterns in complex industrial systems. Finally, the current framework focuses solely on FD, without incorporating multitask learning, which could enable simultaneous predictions of machine states and other related tasks. In future research, we aim to address these limitations through three key steps. First, we will develop adaptive denoising techniques, such as denoising AE or attention-based filters to enhance model robustness in noisy industrial environments. Next, we plan to extend MsDCN by incorporating hierarchical clustering or graph neural networks for diagnosing concurrent faults in complex scenarios. Finally, we intend to integrate FD with prognostics (e.g., remaining useful life prediction) through multitask learning frameworks for broader industrial applications.

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